



Advanced composite wing design for next-generation military UAVs: A progressive numerical optimization framework

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ARTICLE INFO

Article history:

Received 9 November 2024

Received in revised form

21 January 2025

Accepted 20 February 2025

Available online xxx

Keywords:

Aircraft wing

Carbon fiber

Composite

Optimization

UAV

ABSTRACT

The design of unmanned aerial vehicles (UAVs) revolves around the careful selection of materials that are both lightweight and robust. Carbon fiber-reinforced polymer (CFRP) emerged as an ideal option for wing construction, with its mechanical qualities thoroughly investigated. In this study, we developed and optimized a conceptual UAV wing to withstand structural loads by establishing progressive composite stacking sequences, and we conducted a series of experimental characterizations on the resulting material. In the optimization phase, the objective was defined as weight reduction, while the Hashin damage criterion was established as the constraint for the optimization process. The optimization algorithm adaptively monitors regional damage criterion values, implementing necessary adjustments to facilitate the mitigation process in a cost-effective manner. Optimization of the analytical model using Simulia Abaqus™ and a Python-based user-defined sub-routine resulted in a 34.7% reduction in the wing's structural weight after 45 iterative rounds. Then, the custom-developed optimization algorithm was compared with a genetic algorithm optimization. This comparison has demonstrated that, although the genetic algorithm explores numerous possibilities through hybridization, the custom-developed algorithm is more result-oriented and achieves optimization in a reduced number of steps. To validate the structural analysis, test specimens were fabricated from the wing's most critically loaded segment, utilizing the identical stacking sequence employed in the optimization studies. Rigorous mechanical testing revealed unexpectedly high compressive strength, while tensile and bending strengths fell within expected ranges. All observed failure loads remained within the established safety margins, thereby confirming the reliability of the analytical predictions.

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1. Introduction

The importance of UAVs in the aviation sector has increased in recent years. One of the greatest features of these aircraft, which are becoming more prevalent day by day, is their ability to stay airborne for extended periods. Structural weight plays a crucial role in extending their time in the air. Despite significant variations in physical dimensions across different UAV sizes, from small to large, common mission requirements such as endurance and payload capacity remain consistent [1,2]. Extensive research efforts are

underway to develop advanced structural materials with significantly reduced weight. As a result of these studies, carbon fiber reinforced polymer materials are preferred in UAVs [3]. CFRP materials stand out in the engineering field due to their superior properties. Studies have shown that CFRP has a much longer fatigue life compared to other traditional materials. Additionally, the high fracture toughness and energy absorption capacity of CFRP have been highlighted, making it a preferred material in areas requiring reliability and safety, such as the aerospace and automotive industries. Furthermore, it has been noted that CFRP increases the structure's resistance to corrosion, making it an ideal choice for enhancing the durability of structures. These features ensure that CFRP has a wide and valuable range of applications in engineering [4–6]. While exhibiting high specific strength, the anisotropic nature of composite materials, arising from their layered structure,

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Peer review under the responsibility of China Ordnance Society.

<https://doi.org/10.1016/j.dt.2025.02.020>

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necessitates careful design considerations. Key design parameters include laminate stacking sequence and ply thickness. By optimizing these parameters, including varying the number of plies within specific regions, weight reductions of 20%–40% can be achieved depending on the initial component design and the desired performance objectives [7–10].

The lightweight nature and high strength of composite materials contribute to increased fuel efficiency of aircraft, enabling longer flight ranges. In addition, their resistance to corrosion reduces maintenance costs, while their design flexibility allows for the production of aerodynamic and complex-shaped structures [11,12]. Carbon fiber is commonly used in aircraft fuselages, wings, and internal structures due to its lightweight and high strength [11] while aramid fiber is used for ballistic protection and foreign object damage resistance due to its high impact resistance. Glass fiber, known for its cost-effectiveness [13–15], is widely used in components such as body panels and interior panels. Epoxy matrix systems are preferred for their strong bonding properties and resistance to degradation [13,14,16]. The aviation sector aims to leverage these advantages to produce safer, more cost-effective, and structurally robust aircraft [17,18].

Many optimization research in the literature employs approaches including Genetic Algorithms (GAs), Particle Swarm Optimization (PSO), and Differential Evolution (DE) algorithms. These algorithms are extensively utilized to optimize a range of complex problems. Genetic algorithms may show premature convergence with limited population sizes, hence constraining their capacity to attain the global optimal state [1]. PSO may encounter difficulties in identifying global solutions in problems characterized by numerous local minima [2]. Although DE is an effective optimization technique, its performance can be significantly influenced by improper parameter settings, leading to suboptimal results. Recent research has focused on optimizing wing structures by exploring variations in the stacking sequence of composite laminates, while adhering to the principles of classical laminate theory. For instance, Shabeer [19] investigated the impact of incorporating off-axis plies with angles of $\pm 15^\circ$, $\pm 30^\circ$, and $\pm 45^\circ$ into the laminate, deviating from the traditional $0^\circ/90^\circ$ layups. This adjustment yielded a 2% decrease in von Mises stresses and 11% decrease in displacement, demonstrating the efficiency of varying layup angles. An optimal design solution was found by tabulating stress and displacement for each ply combination. In a related study by Mary [20], the scope of composite layup angles was expanded to include a wider range, from 15° to -75° . This broader investigation yielded a significant reduction in von Mises stresses, with a notable 51.7% decrease observed specifically within the 45° – 60° angle range. Moreover, weight optimization studies have explored the effects of altering both the number of layers and layup angles within composite wing structures [21–23]. Optimization efforts involving the modification of the stiffness matrix within finite element analysis have demonstrated significant potential for weight reduction in wing structures. Previous studies have reported weight reductions of up to 26.5% and 21.8% through the implementation of these optimization techniques [24,25]. Furthermore, Fu [26] demonstrated a 19.8% reduction in wing weight through a comprehensive parametric study of composite wing geometries. Subsequent research, employing genetic algorithms for optimization, achieved a 17% weight reduction in composite wing structures [27]. Moreover, significant improvements in wing tip bending stiffness, with a 50% reduction achieved through modifications to the laminate stacking sequence within the wing shell, were reported [28]. These findings collectively emphasize the critical role of optimizing composite layup configurations in achieving enhanced structural efficiency and performance in wing design.

Another study employed genetic algorithms, guided by the Tsai-Wu failure criterion, to optimize the design of a 9.7-m DLR-6F

passenger aircraft wing. Compared to the initial wing design configuration, the optimized design exhibited a significant weight reduction of 29.1%, as reported by Shrivastava et al. [29]. Several studies have observed an increase in wing tip bending as a consequence of weight reduction efforts. This phenomenon suggests a potential decrease in structural rigidity. To address this concern, aeroelastic analyses have been conducted to evaluate the dynamic behavior of lightweight wing structures and ensure that increased bending stiffness does not induce aerodynamic instabilities. These investigations have encompassed both statically and dynamically optimized wing designs.

Upon a comprehensive review of the existing literature, it becomes evident that studies concurrently addressing structural design, the development of a cost-effective, results-oriented custom optimization algorithm based on failure criteria, and experimental validation are notably scarce. In light of this gap, the present study endeavors to contribute to the scientific discourse by undertaking a comprehensive investigation encompassing design refinement, optimization strategies, and experimental validation. This research aimed to enhance the optimal design of UAV wing structures through a two-pronged approach. First, we developed a novel optimization algorithm specifically tailored for UAV wing lightweighting, in collaboration with our aerospace industry partner. Second, the optimized designs were rigorously validated through comprehensive mechanical testing. This novel optimization algorithm, distinct from existing commercial tools, leverages a unique iterative process that builds upon successive design improvements, surpassing the limitations of traditional genetic algorithms. By minimizing the number of iterations, this approach offers a more efficient and effective pathway to achieving optimal design solutions. Material selection and characterization protocols were meticulously tailored to align with the specific operational requirements of the aircraft, guiding the subsequent wing design iterations. Computational Fluid Dynamics (CFD) analyses were employed to ascertain the pertinent loads acting upon the wing geometry. Structural integrity assessments were executed and refined utilizing the Hashin damage criterion, facilitated by Python software for optimization purposes. Finally, the efficiency of the resultant composite layup configurations were experimentally evaluated to substantiate the findings of the study and validate the proposed methodologies.

2. Materials and methods

2.1. Material selection

This research evaluated various fabric types for composite material construction, ultimately selecting carbon fiber reinforced polymers (CFRPs) as the optimal choice due to their exceptional strength-to-weight ratio. Carbon fiber fabric was chosen as the reinforcing material, while epoxy resin was selected as the matrix material. The selection of these materials was guided by the stringent requirements of aviation applications.

As a reinforcement, a unidirectional carbon fiber fabric weighing 300 g per square meter (GSM), manufactured by DowAksa™ and identified by the code UDS300, was selected. For the matrix, LR285 epoxy resin from Hexion™, paired with LH287 hardener liquid (100:40 $\pm$ 2 mixing ratio/parts by weight), was opted for (see Table 1).

2.2. Manufacturing of composite plates

In this study, plates were manufactured for both characterization and validation tests, employing a manual layup method. Carbon fibers were precisely cut to match the plate dimensions, and an

Table 1
Material properties of UDS300 fiber and LR285 resin.

Property	Material	
	UDS300 fiber [30]	LR285 resin [31]
Tensile strength	4900 MPa	70–80 MPa
Compressive strength	—	120–140 MPa
Tensile modulus	245 GPa	3–3.3 GPa
Density	1785.7 kg·m ⁻³	1180–1200 kg·m ⁻³

epoxy-hardener mixture was prepared with a 10% surplus to compensate for epoxy loss during layup. Following this, a flat glass mould was coated with wax and left to dry [32,33]. Prior to applying the epoxy, the mixture underwent a degassing process lasting 15 min in order to eliminate any air bubbles that may have formed during the mixing of the epoxy and hardener. The fabrics have been prepared using a cutting machine (See Fig. S8). Then, an epoxy mixture, was applied to the mould surface, and the fabric was carefully laid within the marked frame. Another layer of epoxy mixture was added on top of the fabric. This process was repeated for each layer, followed by the vacuum bagging process to remove air. A fabric-to-matrix ratio of 1:1 was maintained throughout the lay-up process. To ensure proper consolidation and a smooth surface finish, a flat aluminum plate weighing 80 kg was placed on top of the laminate stack. This applied a uniform pressure of approximately 640 kg/m² (0.064 bar) across the laminate. Following these steps, the plates were cured according to the manufacturer's recommendations. The curing cycle consisted of 24 h at 25 °C followed by 15 h at 80 °C. As a result, the plate production was successfully completed, as shown in Fig. 1.

2.3. Mechanical characterization of composite specimens

Test specimens with dimensions of 15 mm × 250 mm and 25 mm × 250 mm were cut from the manufactured plates in the 0°, 45°, and 90° directions, following the ASTM D3039 standard [34] as shown in Fig. 2(a). To mitigate grip damage during testing, aluminum tabs were bonded to the specimen ends according to the standard. Uniaxial tensile tests were conducted on three specimens for each direction at a crosshead speed of 2 mm/min, as specified in ASTM D3039. Strain measurements were obtained using both a video extensometer for characterization and strain gauges for validation purposes.

In addition, compression test specimens with dimensions of 13 mm × 140 mm were cut from the manufactured plates in the 0° and 90° directions, following the ASTM D6641 standard as

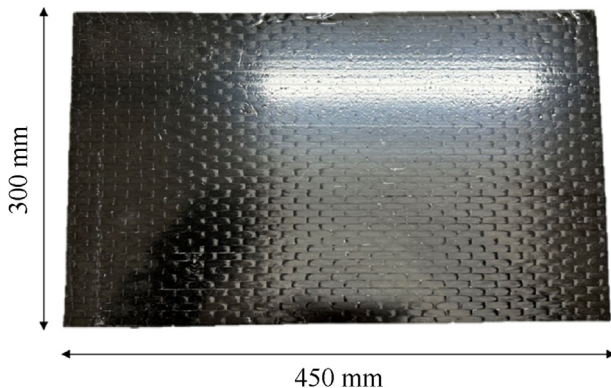


Fig. 1. Manufactured Composite Plate with the lay-up [90°/45°/0°/0°/-45°/90°/45°/0°/-45°] s.

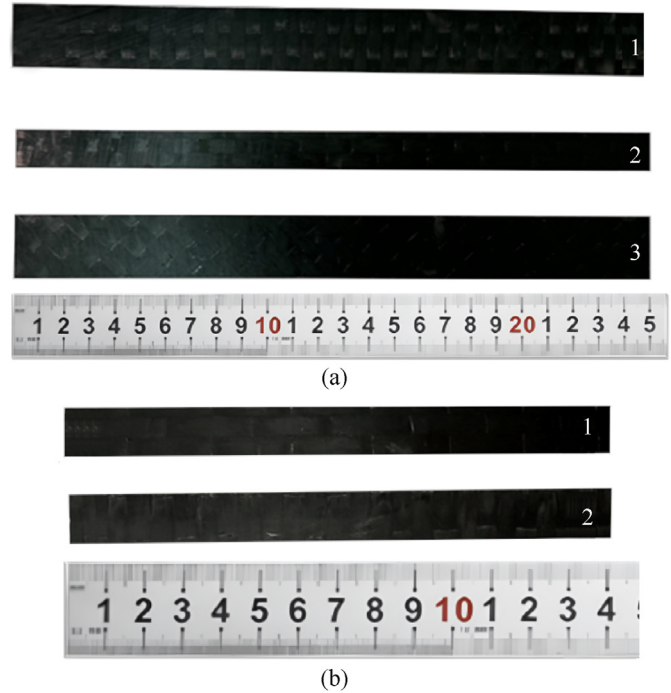


Fig. 2. (a) Tensile test specimens at 90°, 0°, and 45° from top to bottom; (b) Compression test specimens at 0° and 90° from top to bottom.

shown in Fig. 2(b) [35]. Compression tests were conducted on these specimens at a crosshead speed of 1.3 mm/min according to the ASTM D6641 standard.

During the execution of the ASTM 3039, ASTM 6641, and ASTM 2344 tests, the environmental conditions were maintained at 23 ± 2 °C and $50 \pm 10\%$ relative humidity. Tensile and compression tests were performed using an MTS 322.21 model hydraulic tensile test machine and a 100 kN load cell of MTS Model 652 Load Cell Series was used during the tests. During the tensile tests, the machine grips were adjusted for composite tensile specimens, while the combined loading compression (CLC) fixture outlined in the ASTM 6641 standard was employed for compression tests. For strain measurement, Goblet BFLA-B-2-3-3LJC-F strain gauges were utilized.

3. Engineering design and analytical framework

3.1. Aircraft wing design

UAV designs exhibit significant variability, driven by the diverse operational requirements of different missions. This study focuses on the wing design for a military reconnaissance UAV, necessitating a specific mission profile. Key performance parameters for military reconnaissance UAVs include reconnaissance duration and payload capacity.

As illustrated in Fig. 3, the UAV is designed for operation at altitudes ranging from sea level to 6000 m. Key flight performance requirements include a takeoff speed of 42.5 m/s and a cruising speed of 66 m/s. To maximize flight duration and payload capacity, a comprehensive literature review was conducted to identify airfoil profiles with the highest lift-to-drag ratio (C_L/C_D). Based on this analysis, the USA 35B airfoil was selected due to its superior average C_L/C_D performance [36,37].

Based on a comprehensive review of pertinent literature and in alignment with the structural objectives delineated for the chosen

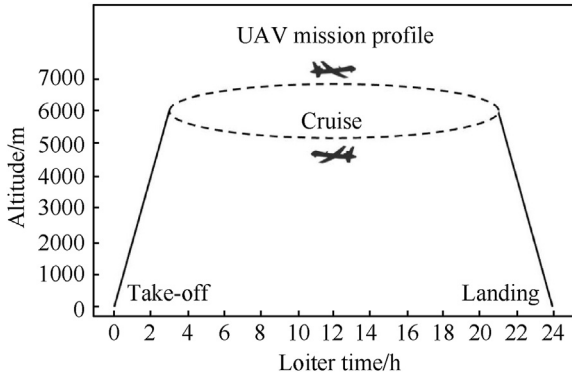


Fig. 3. UAV mission profile.

UAV model, the wing configuration has been determined, as depicted in Fig. 4. The wing length is set at 8 m, with a root chord of 1.2 m and a tip chord of 0.8 m. These dimensions were selected to optimize the UAV's structural integrity and aerodynamic performance, adhering to established design criteria and operational requirements. The aspect ratio of the wing was determined through a comprehensive review of the literature and existing UAV designs, including prominent models such as the TUSAS Anka [38] and the General Atomics MQ-1 Predator [39]. A higher aspect ratio improves endurance through enhanced glide performance and reduced drag, thus expanding the UAV's capacity for prolonged flight. Yet it presents challenges in structural design. Considering these factors and the corresponding research, the dimensions were chosen based on this high aspect ratio [40].

3.2. Structural and aerodynamic load definition

Computational Fluid Dynamics (CFD) analysis was employed to determine the regional loading experienced by the wing geometry under aerodynamic conditions. This analysis accurately predicted the impact of vortices, drag, and other aerodynamic effects on the pressure distribution across the wing surface. The aerodynamic lift force (F_L) and drag force (F_D) are determined by using the critical speed during the take-off and cruise (see Eqs. (1)–(6)). C_L and C_D are obtained based on a reference study conducted by Kermode et al. [41]. Other parameters captured from a Ref. [42] according to the weather conditions in the respective altitude. In CFD simulations, the "far-field" boundary condition represents a region sufficiently distant from the object of interest (e.g., an airfoil) where the influence of the object on the flow field becomes negligible. In essence, the flow in the far-field behaves as if it were in an unbounded domain, unaffected by the presence of the object [43,44].

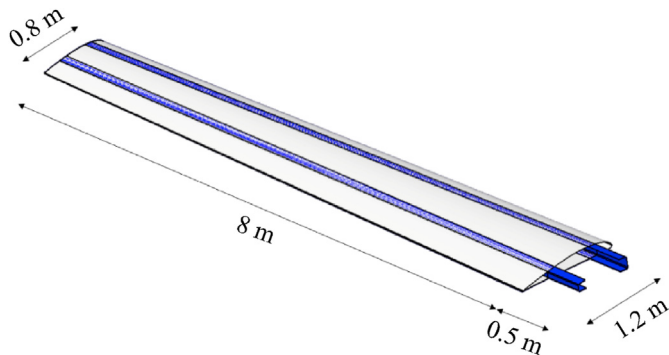


Fig. 4. Wing geometry.

All the terms used in Eqs. (1)–(16) are given in Table 13 in detail.

$$F_L = C_L \cdot \frac{1}{2} \cdot \rho \cdot v^2 \cdot A \quad (1)$$

$$F_D = C_D \cdot \frac{1}{2} \cdot \rho \cdot v^2 \cdot A \quad (2)$$

$$F_{L_{\text{take-off}}} = 1.0569 \times 0.5 \times 1.2251 \times 42.5^2 \times 8 = 9354.2255 \text{ N} \quad (3)$$

$$F_{D_{\text{take-off}}} = 0.0086 \times 0.5 \times 1.2251 \times 42.5^2 \times 8 = 76.12 \text{ N} \quad (4)$$

$$F_{L_{\text{cruise}}} = 0.7312 \times 0.5 \times 0.6601 \times 65^2 \times 8 = 8406.86 \text{ N} \quad (5)$$

$$F_{D_{\text{cruise}}} = 0.0064 \times 0.5 \times 0.6601 \times 65^2 \times 8 = 73.6 \text{ N} \quad (6)$$

The adoption of far-field is important in CFD simulations, particularly for external flow problems such as aerodynamics around ground vehicles or aircraft. In such cases, capturing the flow behavior in the far-field accurately is essential to predict aerodynamic forces and moments [43,44]. The CFD analysis was performed using Ansys SpaceClaim for geometry preparation and Ansys Fluent for numerical simulations. The wing geometry was imported into SpaceClaim and subsequently enclosed within a hemispherical volume. The wing geometry was then subtracted from this volume, resulting in the creation of a computational domain, as illustrated in Fig. 5. The following boundary conditions were then defined for this domain:

- A symmetry condition was applied to the plane of symmetry dividing the hemisphere.
- The wing surface served as the flow surface.
- The outer surface of the hemisphere was designated as the far-field boundary.
- A static pressure of 1 atm was applied to the far-field boundary.

A polyhedral mesh, as depicted in Fig. 6, was generated within the computational domain. Mesh parameters, detailed in Table 2, were carefully considered to ensure accurate flow predictions [45].

The surface mesh comprises 662,064 two-dimensional elements. The maximum element skewness was determined to be 0.577, which falls below the software's recommended limit of 0.6, indicating successful mesh generation [45]. Then, these 2D surface meshes served as the foundation for the creation of a 3D mesh. The resulting 3D mesh comprises 9,062,985 elements with a maximum skewness of 0.897. This value falls below the software's skewness limit of 0.9, ensuring the successful generation of a high-quality 3D

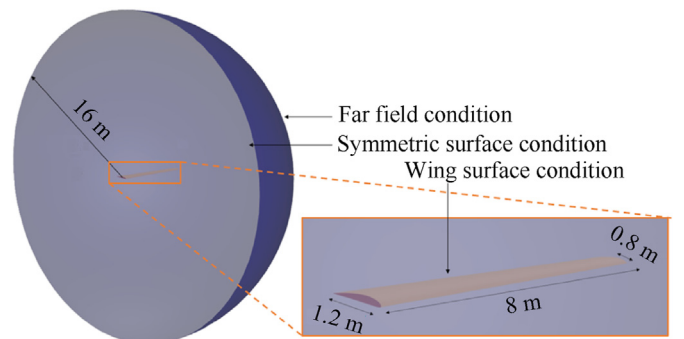


Fig. 5. The wing geometry within the finite volume.

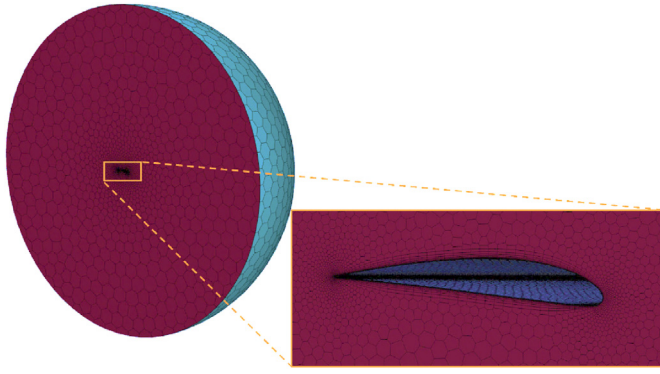


Fig. 6. Computational mesh in polyhedral shape.

Table 2

The regional dimensions of the computational mesh.

Surface	Computation grid size/mm	Growth rate
Wing surface	0.5 mm–40 mm	1.15
Wing closure surface	0.5 mm–10 mm	1.15
Remote surface	0.5 mm–1500 mm	1.15

computational grid (see Fig. S5).

Following these procedures, the computational meshes and boundary conditions were established. The second-order $k-\omega$ SST turbulence model was selected as the turbulence model. This model is widely recognized in the literature as a robust and accurate turbulence model for aerodynamic flows, particularly at low Reynolds numbers [46]. The flow velocity was set to 42.5 m/s with an angle of attack of 4° . The simulation was initialized with a maximum iteration count of 1500, as recommended in Ref. [47] (see Fig. S6).

To minimize distributed pressure on the wing during the structural analysis step, the CFD-post® software was used. The distributed aerodynamic pressures were integrated and represented as equivalent concentrated forces and moments acting at the centroid of each of the ten wing sections.

The sections of the wing are shown in Fig. 7, the force and moments resulting from the CFD analysis are given in Table 3.

3.3. Structural optimization

Finite element analysis (FEA) is an important tool for evaluating the structural integrity of a design under operational conditions. The present research employs computational methods to investigate the structure from a static standpoint, offering valuable insights into its response and structural integrity when subjected to

Table 3

Forces and moments resulting from CFD analysis.

Zone	Location/m			Force/N			Moment/Nm		
	x	y	z	x	y	z	x	y	z
1	0.4	0.6	0	4.299	-41.688	983.992	-186.381	1.084	1.171
2	1.2	0.6	0	4.193	-43.2156	967.117	-178.262	1.846	1.313
3	2	0.6	0	4.207	-43.137	941.78	-168.201	2.210	1.339
4	2.8	0.6	0	4.025	-42.258	911.458	-157.256	2.522	1.288
5	3.6	0.6	0	3.882	-40.628	876.578	-145.640	2.756	1.260
6	4.4	0.6	0	3.702	-38.189	837.101	-133.448	3.078	1.212
7	5.2	0.6	0	3.431	-34.745	791.577	-120.486	3.562	1.189
8	6	0.6	0	3.225	-29.529	734.254	-105.863	4.648	1.220
9	6.8	0.6	0	2.711	-21.188	648.996	-87.553	7.570	1.350
10	7.6	0.6	0	2.125	0.099	496.687	-50.280	17.556	3.551
Total				35.8	-334.478	8189.54	-1333.37	46.832	14.893

Table 4

Elastic properties of materials.

E_{11} /GPa	E_{22} /GPa	ν_{12}	G_{12} /GPa
107 ± 5	7.6 ± 0.4	0.33 ± 0	3.02 ± 0.1

external loads [22,29,48]. The Simulia Abaqus™ 2022 software program was employed for the construction of the finite element analysis model. The structural analysis was initiated by importing the wing geometry into the FE software as surface elements. To accurately represent the distribution of aerodynamic loads, the wing was divided into ten equidistant sections, each spanning 0.8 m. Material properties, as defined in Tables 4 and 5, were assigned to the respective sections within the FE model. For this analysis, a linear elastic material model was assumed, as the material is expected to fail prior to yielding. Predicting the onset and progression of damage in fiber-reinforced composites presents a significant challenge due to the complex interplay of microstructural phenomena. Micromechanical approaches, while valuable, often fall short in accurately capturing these intricate damage mechanisms. These approaches typically require detailed analysis of micro-level stresses, the application of micro-damage criteria to assess damage initiation in individual fibers and the matrix, and subsequently, the prediction of how these micro-damages coalesce to form larger macro-scale damage. To assess damage initiation, this study employs the Hashin damage criteria, utilizing the material properties defined in Table 6. The wing structure was divided into 40 distinct laminate regions. The finite element software must accurately assign the appropriate composite material, fiber orientation, and thickness to each of these regions.

Distinct composite laminates were defined for spars and skin surfaces, as detailed in Tables 6 and 7 [49,50]. Aerodynamic loads, obtained from the CFD analysis at ten discrete points along the wingspan, were applied to the corresponding centroids of the

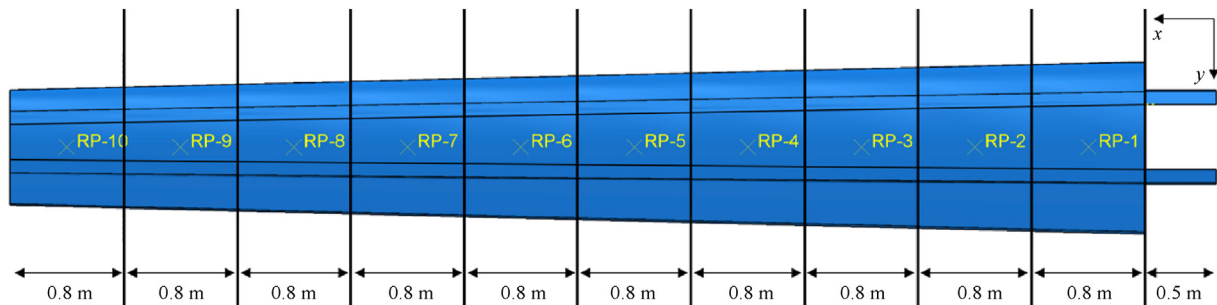


Fig. 7. Sections defined for numerical analysis of the wing.

Table 5

The strength properties of the material.

X_T /MPa	X_C /MPa	Y_T /MPa	Y_C /MPa	S_L /MPa	S_T /MPa
1218 ± 70	356 ± 20	20.1 ± 1	101 ± 6	50.55 ± 3	50.55 ± 3

Table 6

Criterion status.

Criterion status	Modifications
Hashin fiber tensile criterion value > 0.9	Add ply to 0°
Hashin fiber tensile criterion value < 0.7	Remove ply from 0°
Hashin fiber compression criterion value > 0.9	Add ply to 90°
Hashin fiber compression criterion value < 0.7	Remove ply from 90°
Hashin matrix tensile criterion value > 0.9	Add ply to $\pm 45^\circ$
Hashin matrix tensile criterion value < 0.7	Remove ply from $\pm 45^\circ$
Hashin matrix compression criterion value > 0.9	Add ply to 90°
Hashin matrix compression criterion value < 0.7	Remove ply from 90°

Table 7

The distribution of the layup on the wing.

Structural Component	Region									
	1	2	3	4	5	6	7	8	9	10
Top skin	[45°/0°/-45°/90°/45°/0°/-45°]s									
Bottom skin	[45°/0°/-45°/90°/45°/0°/-45°]s									
Front spar	[45°/0°/0°/-45°/90°/45°/0°/0°/-45°]s									
Rear spar	[45°/0°/0°/-45°/90°/45°/0°/0°/-45°]s									

segmented wing geometry. These loads were then evenly distributed across the respective wing surfaces. To constrain the model, the spars were fixed (with six degrees of freedom) along a 0.5-m length at their outboard edges, as illustrated in Fig. 7. A rigid connection was enforced between the spar and skin surfaces at all interfaces, as depicted in Fig. 8. Finally, a computational mesh was generated for the entire wing structure. To simplify the meshing process, the mesh was initially generated on the individual wing surfaces (skin and spar) and subsequently merged into a single, integrated mesh. A two-dimensional surface was used to form the

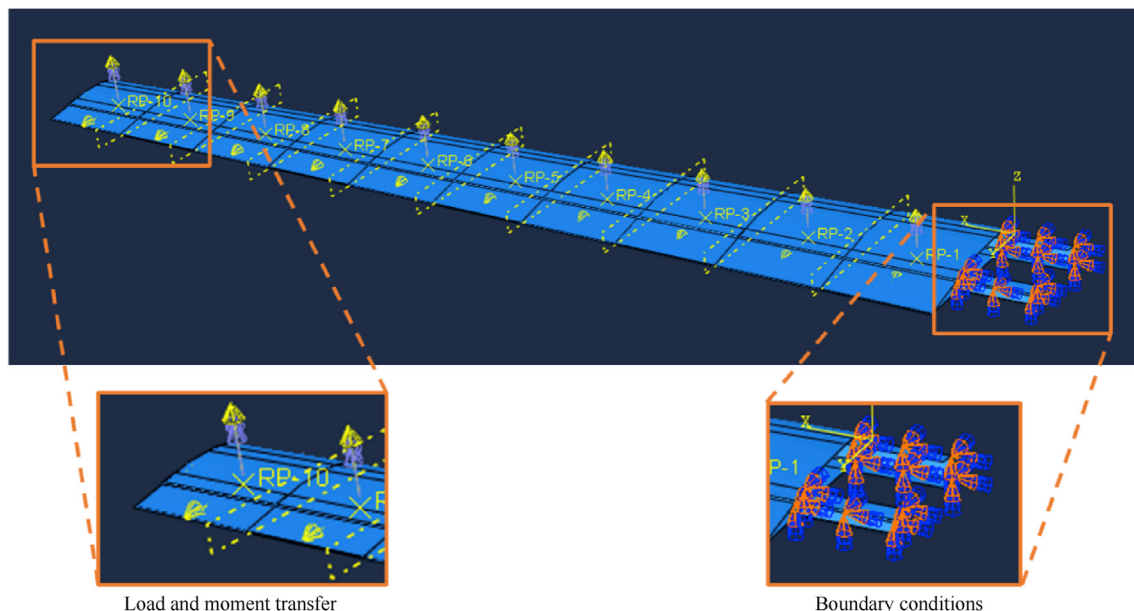
computing grid. The preference for a two-dimensional computing grid stems from its ability to minimize analysis solving times due to the small number of grid nodes, as well as the absence of the requirement for a new computation grid in subsequent laminate updates (see Fig. S7) [49,50]. With this information, a quadrilateral (S4R) computation grid with an element size of 40 mm was generated using the Simulia Abaqus™ 2022 mesh module. The resulting mesh comprises 15,047 elements. Subsequent to these model preparations, the optimization process was conducted using two distinct approaches: a custom-developed algorithm and a genetic algorithm.

3.3.1. Single objective optimization by custom algorithm

The optimization phase has commenced following the completion of FEM analysis, representing a pivotal stage in the entire process. At this juncture, attention is directed towards optimizing the number of layers and their sequential orientations across the two-dimensional structural surfaces. The initial division of these surfaces into 40 discrete partitions requires meticulous adjustment in order to improve the overall structural performance. Table 7 presents information regarding balanced and symmetrically designed laminates that have been developed in accordance with classical lamination theory. It is worth mentioning that the skins included in these laminates exhibit a higher quantity of layers, suggesting their function as load-bearing components.

The optimization procedure is predominantly determined by two critical parameters, both significantly impacting the structural integrity. To begin with, a strategic adjustment is made to the number of plies in order to attain the optimum thickness, thus ensuring the necessary structural strength. Ply orientation ensures strength with minimal plies by directing them at suitable orientation for the loading condition in that region. These optimization strategies are fundamental to achieving enhanced structural performance and minimizing weight. By iteratively adjusting ply count and orientation, the design process iteratively refines the structure towards an optimal solution.

Structural optimization was performed using the Simulia Abaqus™ 2022. An initial structural analysis was conducted, and the results were subsequently used to iteratively refine the design. A

**Fig. 8.** The analysis model of the wing.

custom-developed Python script was integrated within the Abaqus environment to automate the optimization process. This subroutine allows for user-defined input parameters, including the regions to be optimized and the location of the analysis file. A safety factor of 1.25, consistent with industry standards for similar UAV applications, was incorporated into the optimization algorithm to ensure sufficient design robustness. The optimization process involved iteratively modifying the ply orientation within each of the 40 designated regions. Following each design modification, a new finite element analysis was performed, and the results were analyzed to guide the next iteration of the optimization algorithm [51,52].

The optimization code initiates by identifying the corresponding elements within the finite element model for each designated optimization region. Subsequently, an initial finite element analysis is performed using Simulia Abaqus™ 2022. Following the analysis, the code processes the results to generate a report file containing the Hashin failure criteria values for each element within the model. Specifically, the maximum values for Hashin fiber tension, Hashin fiber compression, Hashin matrix tension, and Hashin matrix compression are extracted for each optimization region. These maximum values are then filtered based on a user-defined criterion, as outlined in Eqs. (7)–(14) [53].

Fiber tensile mode:

$$\sigma_{11} > 0 \quad (7)$$

$$F_{ft} = \left(\frac{\sigma_{11}}{X^T} \right)^2 + \alpha \left(\frac{\sigma_{12}}{S^L} \right)^2 = 1 \quad (8)$$

Fiber compression mode:

$$\sigma_{11} < 0 \quad (9)$$

$$F_{fc} = \left(\frac{\sigma_{11}}{X^C} \right)^2 = 1 \quad (10)$$

Matrix tensile mode:

$$\sigma_{22} > 0 \quad (11)$$

$$F_{mt} = \left(\frac{\sigma_{22}}{Y^T} \right)^2 + \left(\frac{\sigma_{12}}{S^L} \right)^2 = 1 \quad (12)$$

Matrix compression mode:

$$\sigma_{22} < 0 \quad (13)$$

$$F_{mc} = \left(\frac{\sigma_{22}}{2S^T} \right)^2 + \left[\left(\frac{Y^C}{2S^T} \right)^2 - 1 \right] \frac{\sigma_{22}}{Y^C} + \alpha \left(\frac{\sigma_{12}}{S^L} \right)^2 = 1 \quad (14)$$

The main purpose of this filter is to add or remove high-strength layers belonging to the effective damage mode criterion which is given in Table 6.

Based on the use of this filter, modifications (ply addition or removal) are implemented, leading to the creation of a novel configuration for the analysis model. The analytical model that has been recently developed and given in Fig. 8 is then solved using Simulia Abaqus™ 2022 and undergoes further reporting and filtering steps. The aforementioned process persists until the layer lay-up reaches a state of equilibrium. Upon completion of the code optimization process, the most appropriate stacking sequence is determined by considering the weight and distribution of Hashin. The system records spatial patterns of Hashin values, layer

configurations, weights, and result files for every stage of analysis. Fig. 9 illustrates the schematic depiction of the optimization process.

This study employs a single-objective optimization approach to minimize the weight of the wing structure. A custom script utilizing the Part Simulation technique within the Abaqus environment was developed to achieve this objective. Single-objective optimization aims to optimize a specific design parameter, in this case, the wing weight. This optimization problem can be mathematically formulated as shown in Eqs. (15) and (16):

$$\text{Objective : } \min \max f(x) \quad (15)$$

$$\text{Constraint : } g_i(x) \leq 0, \quad h_j(x) = 0 \quad (16)$$

here, $f(x)$ is the objective function that needs to be optimized, while $g_i(x)$ and $h_j(x)$ represent the constraints in the problem. Table 8 provides various parameters used in this optimization study.

3.3.2. Genetic algorithm development for single objective optimization

Genetic algorithms (GAs) are powerful computational optimization techniques inspired by the principles of natural selection and genetic evolution. They iteratively refine a population of potential solutions by applying operators that mimic natural processes such as selection, crossover, and mutation. GAs have proven effective in solving a wide range of optimization problems across various domains and continue to be an active area of research. Fig. 10 presents a basic flowchart illustrating the key steps involved in a typical GA.

The primary objective of this work is to reduce the weight of the wing structure within specified restrictions; hence, the fitness function was established in Eq. (17).

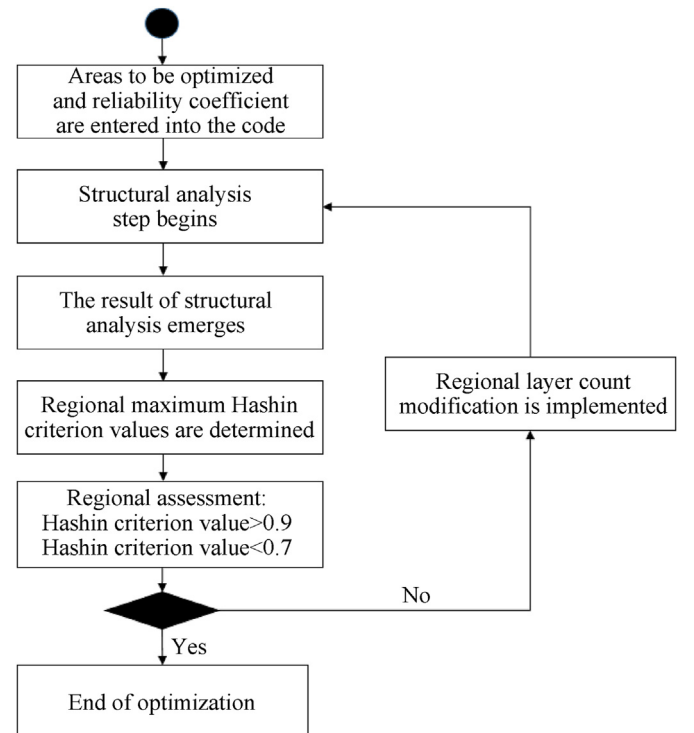


Fig. 9. Optimization schema.

Table 8
Optimization parameters.

Parameter type	Parameter	Value
Design variable	Ply angle	0°, 90°, 45° ve -45°
Design variable	Number of ply	—
Constraint	Hashin criteria	Less than 0.8
Constraint	Stacking sequence	Symmetry
Constraint	Stacking sequence	45° ve -45° balance
Constraint	Number of ply	More than 3
Objective	Minimize weight	—
Design response	Static analysis	—

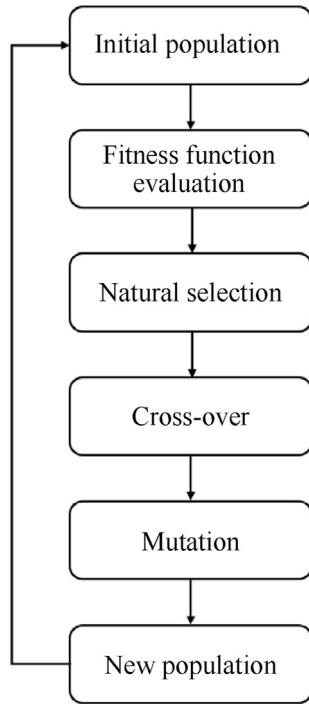


Fig. 10. Flowchart of the simple genetic algorithm [54].

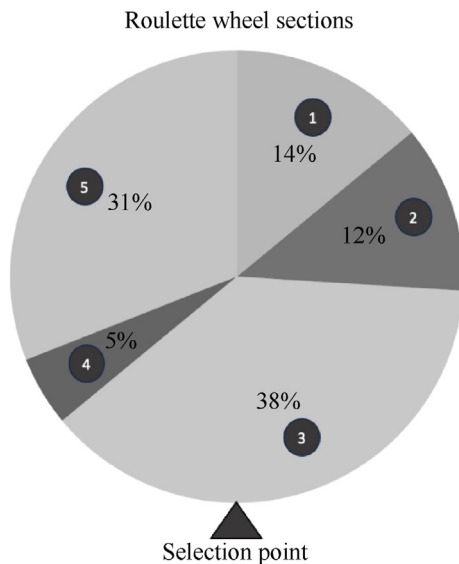


Fig. 11. Roulette wheel selection method.

$$\text{Fitness function} = \frac{w}{f} \quad (17)$$

where w is the weight factor of f , and f is the first selection criterion that considers the static strength of the design solution in terms of Hashin failure criteria.

A variety of selection methods are employed in genetic algorithms, such as tournament selection, rank-based selection, and roulette wheel selection. This study utilizes roulette wheel selection due to its simplicity and computational efficiency, making it a suitable choice for the current optimization problem. A graphical representation of the roulette wheel selection process is provided in Fig. 11 [55,56].

A single-point crossover operator was employed in this study. This operator selects a random crossover point along the chromosome and exchanges genetic material between the two parent chromosomes at that point. Mutation is a critical operator in genetic algorithms, introducing genetic diversity and preventing premature convergence towards suboptimal solutions. To enhance the robustness of the optimization process, an adaptive mutation strategy was implemented. By dynamically adjusting the mutation rate during the optimization process, this approach helps to maintain exploration of the search space while avoiding stagnation in local optima [57,58].

To enhance computational efficiency, an adaptive mutation strategy was implemented with a maximum mutation probability of 25%. The Adaptive Mutation Probability (AMP) was calculated as follows:

$$AMP = P_{\max} \left(W_N \frac{N_0 P_n}{N_0 P_t} \right) \quad (18)$$

where $P_{\max}$ is the maximum mutation probability (25%), W_N is the weight factor for each design region, $N_0 P_n$ is the number of plies at related zone, and $N_0 P_t$ is the total number of plies of the component. This formula adjusts mutation probability for each region dynamically.

The mutation process focuses on modifying or deleting plies in the design regions, adding diversity and refining the design solution. After mutation operations are performed, a new generation of individuals is created, guiding the algorithm toward an optimal solution while maintaining population diversity.

4. Results and discussions

4.1. Mechanical characterization of composite specimens

The mechanical properties of carbon fiber reinforced polymers (CFRPs) are highly dependent on the specific combination of fiber and matrix materials. Therefore, prior to the design and analysis, it was crucial to experimentally characterize the strength and elastic properties of the CFRP material utilized in this study.

Following manufacturing, uniaxial tensile and compression tests were conducted on specimens cut at 0°, 90°, and 45° orientations, in accordance with ASTM D3039. The experimental results, summarized in Tables 4 and 5, provide the measured mechanical properties of the fabricated CFRP material.

The experimentally determined mechanical properties of the CFRP material were subsequently incorporated into the finite element models for subsequent structural analysis and optimization.

4.2. Optimization results

4.2.1. Analysis of optimization results from a custom-developed algorithm

The optimization process involved 45 iterative analysis steps. Convergence of the optimization process was achieved when both the Hashin damage criteria values and the structural weight stabilized within acceptable tolerances.

Fig. 12 illustrates the evolution of the maximum Hashin failure criterion value across all optimization iterations. Initially, the maximum Hashin value tends to increase as the optimization algorithm reduces the number of plies. Subsequently, the values converge as further ply reductions are implemented. A detailed analysis of the final optimization stages revealed that the removal or addition of single or double plies had a significant impact on the maximum Hashin value. By carefully evaluating these design variations, the optimization algorithm identifies the optimal ply configuration that maximizes performance while minimizing weight.

Fig. 12 depicts the relationship between weight reduction and the evolution of the maximum Hashin failure criterion value across successive optimization iterations. As expected, a general trend of decreasing weight is observed with increasing Hashin values, consistent with findings reported in the Ref. [29]. The optimization algorithm aims to minimize weight while maintaining a safety margin by constraining the maximum Hashin value to a threshold of 0.8. The optimization process iteratively reduces the number of plies within the structure, converging towards the target Hashin value. If convergence is not achieved within the specified number of iterations, the optimization process is terminated. Upon reaching the desired convergence criteria, the algorithm identifies the lightest structural configuration that satisfies the specified Hashin constraint, representing the optimal design.

The optimization process yielded a variety of laminate stacking sequences. Analysis revealed that several iterations resulted in designs with Hashin failure criteria values approaching the specified limit of 0.8. The 16th iteration demonstrated the lowest structural weight, achieving a significant reduction from 52.7 kg to 34.4 kg, representing a weight reduction of 34.7%. Table 9 presents the optimized laminate stacking sequences for each wing component as determined in the 16th iteration.

An increase in cross-sectional thickness is observed towards the wing root, where the structure is fixed (see Table 9). This design trend is consistent with the expected behavior of a wing under bending loads. The optimization process resulted in thicker spar sections compared to the skin panels, indicating that the spars are

primarily responsible for carrying the majority of the bending loads.

Two key design variables were considered in this optimization study: ply count and ply orientation, as summarized in Table 8. Analysis revealed that while both variables influence the Hashin failure criterion, ply orientation exerts a more significant impact on the convergence behavior of the optimization process. Fig. 12 illustrates that substantial variations in Hashin are associated with changes in the number of plies. On the other hand, the regions on the wing with minimal variations in Hashin are predominantly affected by ply orientation. The effect of ply orientation on the Hashin damage criterion value ranges from 5% to 15%, while alterations in the number of plies can lead to variations of 20%–40%. It is essential to recognize that these values may vary based on the size, thickness, and loading conditions of the region on the wing.

Analysis of the optimized design, as summarized in Table 10, reveals several key observations. While the tensile stresses within the wing components remained relatively unchanged, a significant increase in compressive stress (σ_{11}) was observed, rising from 130 MPa to 317.8 MPa. This increase in compressive stress resulted in the maximum compressive stress in the fibers approaching the Hashin failure criterion limit of 0.8. The reduced stresses found in alternative orientations can be attributed to the bending encountered by the wing structure when subjected to the imposed load conditions. In the context of composite design, it is anticipated that the major direction of the load would be positioned above the 0° orientation. This observation is indicative of the load being applied from the appropriate direction, as demonstrated by the obtained results. Furthermore, the optimization process resulted in an increase in wingtip deflection, from 303.4 mm to 575.1 mm. This increase in deflection is a direct consequence of the weight reduction efforts, which inherently impact the structural stiffness and the load-bearing performance of the wing. The initial design utilized a uniform composite layup across the entire wingspan, which may have contributed to the significant weight reduction.

Analysis of the optimization results, summarized in Table 10, identified Region 1 of the front spar as the most critical region. The optimization process resulted in an increase in compressive stresses within this region, leading to an increase in the Hashin compressive damage criterion value towards the critical limit of 0.8. Conversely, tensile stresses in this region exhibited a slight decrease. These stress changes are further illustrated in Fig. S3. The change ratio is indicated with a (+) sign for increases and a (–) sign for decreases in the post-optimization values compared to the pre-optimization values. To validate the analytical predictions, a composite plate representative of Region 1 was fabricated using the hand lay-up technique, as previously described. This test specimen was subsequently subjected to a series of structural tests.

4.2.2. Optimization results implemented by genetic algorithm

In this section, structural optimization was performed using a genetic algorithm, incorporating the design constraints and variables outlined in Table 8. The optimization process was conducted with three distinct initial population sizes: 8, 16, and 32 individuals. Hybridization was performed within each population. To enhance population diversity and facilitate effective hybridization, the initial population was not restricted to a specific configuration.

Fig. 13 illustrates the evolution of structural weight across successive generations of the genetic algorithm. A consistent trend of weight reduction is observed, accompanied by corresponding adjustments in the Hashin damage criterion. As iterations advance, this trend becomes more pronounced.

Fig. 13 demonstrates that an initial population of 8 indicates that eight different configurations are evaluated per generation. Consequently, the 16 and 32 population scenarios require longer

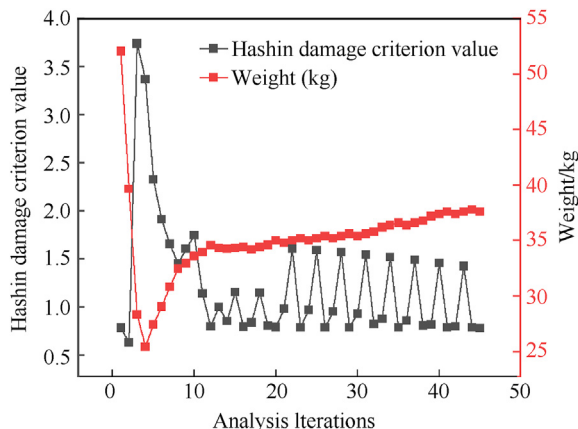


Fig. 12. Hashin damage criterion value and weight changes per iteration.

Table 9
Stacking Sequences in wing components at 16th iteration.

Region	Top skin layer orientation	Bottom skin layer orientation	Front spar layer orientation	Rear spar layer orientation
1	[45°/0°/-45°/90°/45°/0°/-45°]s	[45°/0°/0°/90°/0°/0°/-45°]s	[90°/45°/0°/0°/-45°/90°/45°/0°/-45°]s	[45°/0°/0°/-45°/90°/45°/0°/-45°]s
2	[45°/0°/90°]s	[45°/0°]s	[-45°/0°/0°/45°/0°]s	[45°/0°/-45°/0°]s
3	[45°/0°]s	[45°/0°]s	[45°/0°/0°/-45°]s	[45°/0°/-45°/0°]s
4	[45°/0°]s	[45°/0°]s	[45°/0°/-45°]s	[45°/0°/-45°/90°]s
5	[45°/0°]s	[45°/0°]s	[45°/0°/-45°]s	[45°/0°/-45°]s
6	[45°/0°]s	[45°/0°]s	[45°/0°/-45°]s	[45°/0°/-45°]s
7	[45°/0°]s	[45°/0°]s	[45°/0°/-45°]s	[45°/0°/-45°]s
8	[45°/0°]s	[45°/0°]s	[45°/0°/-45°]s	[45°/0°/-45°]s
9	[45°/0°]s	[45°/0°]s	[45°/0°/-45°]s	[45°/0°/-45°]s
10	[45°/0°]s	[45°/0°]s	[45°/0°/-45°]s	[45°/0°/-45°]s

Table 10
The assessment of the analysis.

Parameter	Pre-optimization	Post-optimization	Change ratio
σ_{11}	298.3 MPa	308.0 MPa	+3.3%
	-130.1 MPa	-317.8 MPa	+143.8%
σ_{22}	17.1 MPa	16.5 MPa	-3.5%
	-21.3 MPa	-22.3 MPa	+4.7%
σ_{12}	13.0 MPa	12.0 MPa	-7.6%
	-13.4 MPa	-19.1 MPa	+42.5%
Bending	303.4 mm	575.1 mm	+89.7%
Maximum Hashin fiber compression criterion	0.1336	0.797	+491%
Maximum Hashin fiber tensile criterion	0.0774	0.0639	-17.4%
Maximum Hashin matrix compression criterion	0.0803	0.153	+90.5%
Maximum Hashin matrix tensile criterion	0.786	0.694	-11.7%
Weight	52.7 kg	34.4 kg	-34.7%

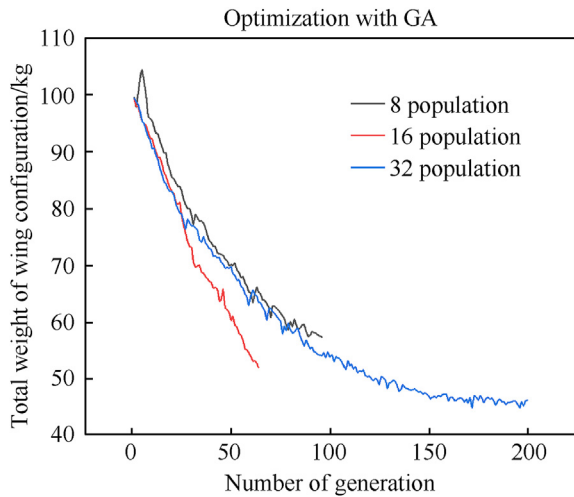


Fig. 13. Weight changes per generation with GA.

generation times. However, despite the increased computation time, larger populations enhance diversity and improve hybridization, thereby increasing the likelihood of converging toward the optimal solution. In contrast, the rate of weight reduction in the 8 and 16-state scenarios remained relatively lower due to limited diversity compared to the 32-state scenario. The configuration with a population size of 32 proved to be the most effective, achieving a substantial weight reduction of 55% by the end of the 201th generation, with the initial weight of 99.5 kg decreasing to a final weight of 44.7 kg.

Table 11 presents the lay-up of the minimum-weight configuration obtained through optimization with GA. A comparison of the custom script and the genetic algorithm reveals distinct advantages for each method. The custom script exhibits higher computational

efficiency, achieving weight optimization with fewer analysis steps. Conversely, the genetic algorithm, while requiring more computational time due to crossover and population diversity, explores a broader solution space, potentially leading to a more globally optimal design. The custom script prioritizes rapid identification of an effective solution, while the genetic algorithm leverages population diversity to explore a wider range of design possibilities.

Compared to the custom script, the genetic algorithm (GA) initially utilized a larger number of plies. This strategy was intended to increase population diversity and prevent premature convergence during the optimization process. As shown in Fig. 13, this approach facilitated a greater reduction in weight. However, in scenarios where the primary goal is to identify the lightest layout, rather than maximizing the proportional weight reduction from the initial design, the custom script emerges as a more effective optimization method. Although the GA achieved a more substantial weight reduction relative to the baseline, its performance was hindered by slowdowns and interruptions during the hybridization and crossover stages. In contrast, the custom script evaluates each layer independently, unaffected by genetic operator probabilities. This characteristic allows for faster convergence, requiring fewer iterations. As a result, the custom script produced a design that was 10.3 kg lighter than the lightest configuration found by the GA.

In this work, we utilized the tailored optimization script over 40 distinct regions of the wing. This script concurrently assesses the Hashin damage criterion across all regions by adding or removing plies when needed for each. The optimization performed with the script was 133 times faster than that conducted using a genetic algorithm and achieved an additional 23% reduction in weight. As the number of regions and wing dimensions increase, this difference grows exponentially. Consequently, it becomes feasible to optimize all regions concurrently, even when their size escalates. In extensive structural optimization investigations, the runtime will increase based on convergence criteria, although the analysis steps will not demonstrate exponential growth. The quantity of areas

Table 11
Stacking sequences in wing components after GA.

Region	Top skin layer orientation	Bottom skin layer orientation	Front spar layer orientation	Rear spar layer orientation
1	[90°/0°/90°/45°/-45°/0°/90°/0°/45°/-45°/0°/0°/0°/45°/-45°/90°/0°/45°/-45°]s	[0°/0°/0°/0°/0°/0°/0°/0°/45°/-45°/45°/-45°/45°/-45°/0°/0°/0°/0°/90°/0°/0°/90°/0°]s	[45°/-45°/90°/90°/0°/0°/45°/-45°/0°/0°/0°/45°/-45°/0°/0°/45°/-45°/0°/0°/0°]s	[0°/0°/0°/45°/-45°/45°/-45°/0°/45°/-45°/0°/0°/0°/90°/0°/0°/90°]s
2	[90°/0°/0°/0°/45°/-45°/45°/-45°/45°/-45°/90°/0°/45°/-45°]s	[0°/0°/0°/0°/0°/0°/90°/0°/0°/0°/0°/0°/0°/90°/0°]s	[0°/0°/0°/0°/45°/-45°/90°/0°]s	[0°/0°/90°/90°/0°/0°/45°/-45°/45°/-45°]s
3	[90°/0°/0°]s	[0°/0°/90°/90°/0°/0°/0°/0°]s	[0°/90°/0°]s	[0°/0°/0°]s
4	[45°/-45°/90°/90°/45°/-45°/0°]s	[0°/90°/45°/-45°/0°]s	[90°/90°/90°]s	[45°/-45°/90°]s
5	[0°/90°/0°]s	[45°/-45°/45°/-45°/0°/0°/-0°]s	[90°/90°/0°]s	[0°/0°/0°]s
6	[90°/0°/0°]s	[0°/90°/0°/45°/-45°/0°/45°/-45°]s	[90°/0°/0°]s	[90°/0°/0°]s
7	[90°/0°/0°]s	[45°/-45°/45°/-45°/45°/-45°/0°]s	[90°/0°/90°]s	[0°/0°/0°]s
8	[45°/-45°]s	[45°/-45°/0°/0°/90°/45°/-45°]s	[0°/0°/0°]s	[45°/-45°]s
9	[45°/-45°/0°]s	[90°/45°/-45°/45°/-45°]s	[0°/0°/0°]s	[90°/0°/90°]s
10	[0°/0°/0°]s	[45°/-45°/90°/45°/-45°]s	[90°/45°/-45°]s	[0°/45°/-45°]s

increases linearly with the number of iteration steps, regardless of the time required to resolve the issue. This methodology necessitates far fewer solution iterations compared to methods like genetic algorithms, as it does not depend on probabilistic operations. This efficiency underscores a significant benefit of the tailored script. Upon examining the final results, the custom script provides a more optimized solution. Therefore, the experimental verification phase proceeded using the configuration provided by the custom script.

4.3. Verification of numerical results through experimental methods

Region 1 of the front spar, identified as the most critical area for structural integrity, was selected for experimental validation. The thickness increases the load capacity and also ensures a solid connection with the body. The failure of this region is more critical for the wing compared to other areas. Here a comparative analysis was performed on a test sample using tensile tests and FE analysis having 18-layer with [90°/45°/0°/-45°/90°/45°/-45°]s layup sequence. The tests were performed in accordance with ASTM D3039 standards (see Figs. S1 and S2). Then, the tests were compared using a tensile test model through FE analysis on Abaqus as shown in Fig. 14(a). When subjected to an applied load of 99.5 kN, the Hashin matrix tension criterion attains the critical

threshold. However, the specimen of same dimensions was subjected to testing in the test apparatus, revealed an average fracture value of 91 kN. Fig. 14(b) shows an image of a fractured tensile test specimen. Upon examination of this figure, it is understood that the fracture mode of the failed specimen corresponds to the fracture type coded as DGM shared within the scope of the standard. In Fig. 14(c), the fracture type code DGM indicates that the letter D represents edge delamination, the letter G represents fracture within the gauge area, and the letter M represents mid-span fracture.

Following the completion of tests in accordance with ASTM D6641, a numerical analysis was performed on the specimens subjected to compressive loads utilizing Simulia Abaqus™ 2022. The investigation demonstrated that the Hashin matrix compression value of the material attained the threshold value (as illustrated in Fig. 15(a) with red contours) under a load of 23 kN. In the experiments conducted under identical conditions, the sample exhibited fracture at an average load of 24.8 kN. The fracture occurred between the fixtures, as illustrated in Fig. 15(b), thereby confirming the test's compliance with the established requirements. A comparison of the analytical simulation image and the experimental result indicates that fracture initiates near the grips, displaying a similar pattern.

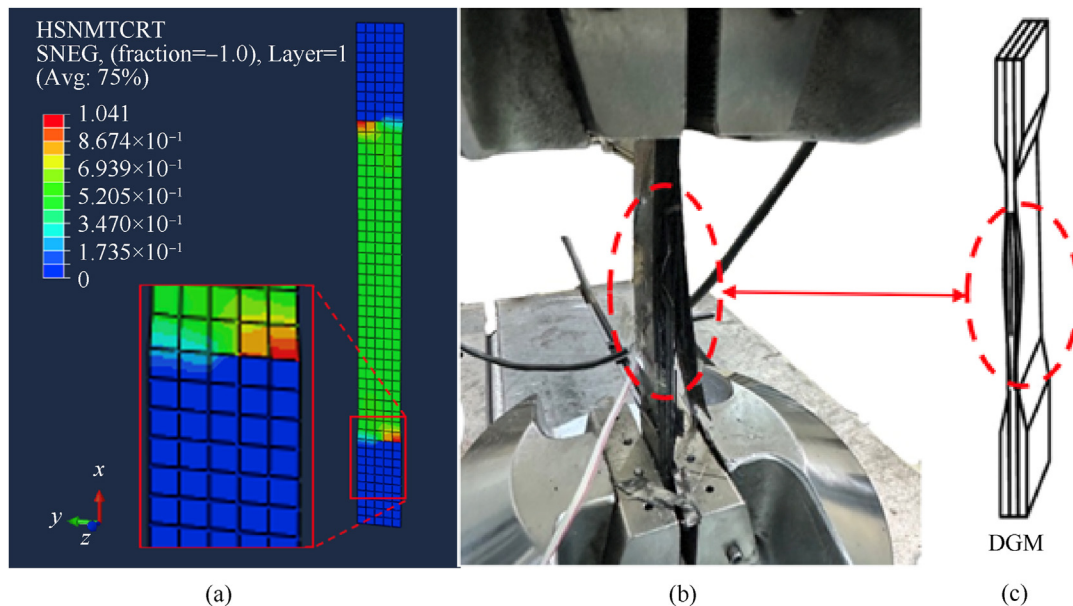


Fig. 14. (a) Tensile test Abaqus™ simulation; (b) Tensile test; (c) Fracture type found in the ASTM 3039 standard [34].

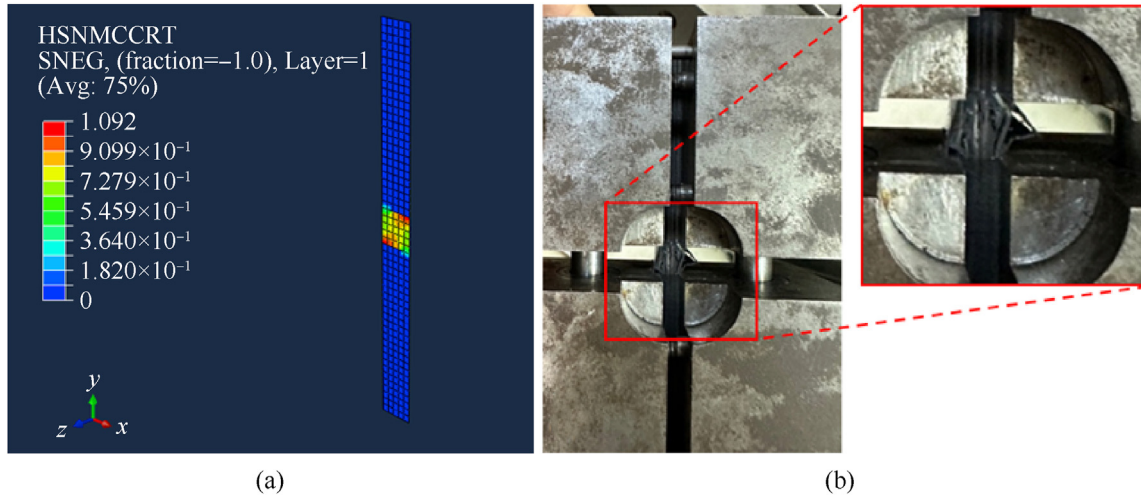


Fig. 15. (a) Compression test Abaqus™ simulation; (b) Compression test.

Table 12
Test results.

Test Type	Specimen orientation	Test $\sigma_{fracture}$ / MPa	FEM simulation $\sigma_{fracture}$ / MPa	Error rate
Uniaxial tensile test	0°	515 ± 8	571	9.8%
Uniaxial tensile test	90°	235 ± 7	253	7.1%
Compressive strength test	0°	311 ± 21	289	7.6%
Compressive strength test	90°	261 ± 4	253	3.1%

Table 13
List of parameters.

Symbol	Parameter Description
F_L	Lift force
F_D	Drag force
C_L	Lift coefficient
C_D	Drag coefficient
ρ	Density
v	Velocity
A	Wing area
$F_{Ltake-off}$	Lift force at take-off
$F_{Dtake-off}$	Drag force at take-off
$F_{Lcruise}$	Lift force at cruise
$F_{Dcruise}$	Drag force at cruise
F_{ft}	Hashin value at fiber tensile mode
F_{fc}	Hashin value at fiber compression mode
F_{mt}	Hashin value at matrix tensile mode
F_{mc}	Hashin value at matrix compression mode
σ_{11}	Stress tensor at 11 (x) direction
σ_{12}	Stress tensor at 12 (xy) direction
σ_{22}	Stress tensor at 22 (y) direction
X^T	Longitudinal tensile strength
X^C	Longitudinal compressive strength
Y^T	Transverse tensile strength
Y^C	Transverse compressive strength
S^L	Longitudinal shear strength
S^T	Transverse shear strength
α	Angle of fiber

Table 12 presents a comparison of the experimental and analytical results, highlighting the observed error levels. The analysis indicates that the error levels fall within acceptable limits for the given application [59–61].

Experimental tensile tests exhibited slightly lower strength values compared to the FEA predictions, while compression tests demonstrated higher experimental strengths. These discrepancies

may be attributed to variations in specimen thickness, as reported in Ref. [59]. The performed experiments aimed to verify the technique of structural analysis modeling, thereby offering critical insight into the accuracy and dependability of the analytical calculations. Additionally, it is significant to mention that the Compression Test (with respect to ASTM D6641 standard) is designed for symmetric and balanced layups, and since it involves compression in one direction with the aid of a pinned tool, buckling effects can be considered negligible, close to zero.

Lastly, upon seeing the delaminations depicted in Fig. 14(b), a decision was made to assess the interfacial strength of the fabricated laminate composites. We fabricated interfacial shear strength (ILSS) test samples with 0° fiber angle following the ASTM D2344 testing standard and performed tests on three samples utilizing the Shimadzu AG-x universal testing machine with a 50 kN load cell. Throughout this experiment, a rectangular beam-type specimen with dimensions 41 mm × 13.6 mm × 6.9 mm was subjected to bending load using a three-point bending fixture with a crosshead speed of 1.3 mm/min. A point load is applied in the exact center of the beam, which is supported on both sides. The act of bending causes an uneven distribution of stresses throughout the cross-section of the beam. The maximum shear stress is located near the centerline of the beam, precisely beneath the place where the load is applied. During a well-executed short beam strength (SBS) test for interlaminar shear strength (ILSS), the shear stress in the mid-plane surpasses the normal bending stresses and becomes the primary stress. This is accomplished by maintaining a support span that is relatively narrow in comparison to the thickness of the beam. As the applied force rises, the composite material predominantly breaks apart because the shear stress surpasses the interlaminar shear strength between the layers. Delamination within the composite is a common indicator of this failure (see Fig. 16 and Fig. S4). It is important to mention that the SBS test does not directly apply pure shear stress. Instead, it generates a bending

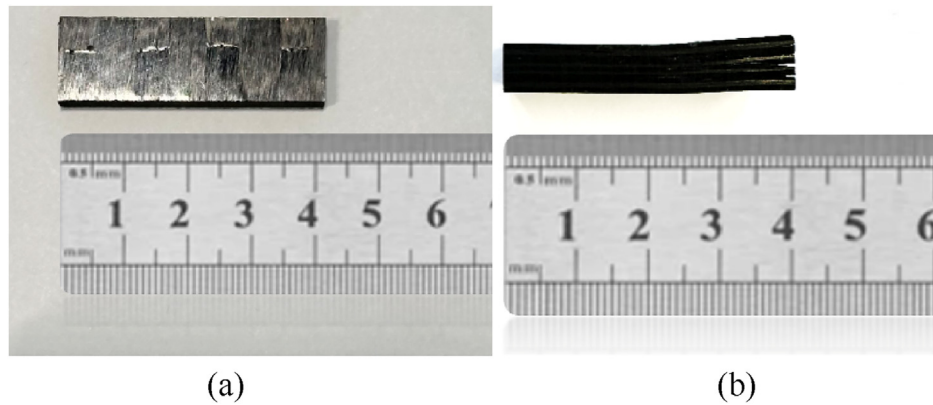


Fig. 16. (a) Specimen before the ILSS Test; (b) Specimen after the ILSS test.

condition where shear stress becomes the primary cause of failure.

The test findings indicate that the short beam strength value is 29 ± 7 MPa. By comparing this number with the interlaminar shear strength catalog value of 47 MPa for carbon fiber reinforcement in the LR285 Hexion™ matrix, we can conclude that the material's mechanical performance did not meet the expected values. This can be attributed to the production process, namely the hand-layup method employed to produce the test samples. These challenges can be resolved by employing more sophisticated production procedures, such as VARTM or pre-preg techniques.

5. Conclusions

The study aimed to develop optimized composite structure in order to have less structural weight for better performance and longer endurance according to mission profile. The present investigation started with a thorough production and characterization of the composite material planned for the primary target structure which is a specific UAV wing. Subsequently, an aerodynamically sound wing shape was designed. We performed thorough structural analysis and utilized a Python script in Simulia Abaqus™ 2022 to iteratively optimize the wing structure, therefore ensuring its structural integrity under flight loads. The single objective optimization script is uniquely developed for this study without using optimization algorithms available in the market. The newly developed optimization algorithm is not based on crossing the design variables based on genetic algorithms, instead this optimization technique improves the objective based on each preceding step result until satisfying the design objective and constraints. Hence, this capability of the algorithm introduces a notable novelty by achieving the optimal design solution in a reduced number of iterations. The improved optimization method resulted in a significant decrease of 34.7% in structural weight which supports the main aim of this study. The optimization method and structural improvement developed in this study are of a nature that will also serve as a guide for other studies. Furthermore, the optimization algorithm developed specifically for this study has yielded better results than existing optimization algorithms, thus proving that custom-developed optimization methods can be more efficient. No matter how high the strength of CFRP material is, it has been shown that the structure's stacking can be much more efficient with the developed optimization method, thereby ensuring the most optimal use of the material. The manufactured specimens from the most critical segment of the optimized wing structures were subjected to comprehensive testing, wherein the obtained findings were compared and validated with the structural analysis results.

The custom script developed in this study is limited to the predefined parameters for optimization. It operates within these defined constraints and is restricted to static analysis. In future studies, the script could be extended to include additional load cases, allowing simultaneous static and dynamic analyses. Beyond the damage criteria, constraints such as tip deflection and buckling could also be incorporated. In addition, it could be transformed into a multi-objective optimization code, considering objectives like weight minimization alongside load-carrying capacity.

Replication of results

The data used to support the findings of this study are available from the corresponding author upon request. The authors state that the paper contains all information necessary to reproduce the results.

CRediT authorship contribution statement

M. Atif Yilmaz: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kemal Hasirci:** Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Berk Gündüz:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alaeddin Burak Irez:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of interest statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to Dr. Korhan Sahin from TUSAS Aerospace for their valuable technical comments on the implementation of the proposed material for UAV wings. We kindly thank Muhammed Yusuf Yasar, Asya Nur Sunmaz, and Samet Sahin for their assistance during CFD analysis. We sincerely appreciate Muslum Cakir for his assistance in the experimental characterization of this study. This research was supported by the Istanbul Technical University Office of Scientific Research Projects

(ITUBAPSIS), under grant MYL-2022-43776. We acknowledge the use of AI-based tools for language editing and enhancement in preparing this manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.dt.2025.02.020>.

References

- [1] Goldberg DE, Holland JH. Genetic algorithms in search, optimization, and machine learning. *Mach Learn* 1979;3:95–9.
- [2] Kennedy J, Eberhart R. Particle swarm optimization. *Proc ICNN'95 - Int Conf Neural Networks*; vol. 4:1942–1948. <https://doi.org/10.1109/ICNN.1995.488968>.
- [3] Seyhan A, Irez AB, Polat Y. Development of a polyamide 6-based composite material for UAV propellers. *Conf Proc Soc Exp Mech Ser* 2023;73–7. https://doi.org/10.1007/978-3-031-17445-2_10/TABLES/5.
- [4] Li C, Xian G, Li H. Effect of postcuring immersed in water under hydraulic pressure on fatigue performance of large-diameter pultruded carbon/glass hybrid rod. *Fatig Fract Eng Mater Struct* 2019;42:1148–60. <https://doi.org/10.1111/FFE.12978>.
- [5] Lal HM, Uthaman A, Li C, Xian G, Thomas S. Combined effects of cyclic/sustained bending loading and water immersion on the interface shear strength of carbon/glass fiber reinforced polymer hybrid rods for bridge cable. *Constr Build Mater* 2022;314:125587. <https://doi.org/10.1016/J.CONBUILDMAT.2021.125587>.
- [6] Guo R, Xian G, Li C, Hong B. Effect of fiber hybrid mode on the tension–tension fatigue performance for the pultruded carbon/glass fiber reinforced polymer composite rod. *Eng Fract Mech* 2022;260:108208. <https://doi.org/10.1016/J.ENGFRACMECH.2021.108208>.
- [7] Kuznetsov I, Kozhin V, Novokoshenov A, Elsen T, Nemov A. Optimization approach for the core structure of a wind turbine blade. *Struct Multidiscip Optim* 2024;67:1–16. <https://doi.org/10.1007/S00158-024-03812-Z/METRICS>.
- [8] Hermansen SM, Lund E. Multi-material and thickness optimization of a wind turbine blade root section. *Struct Multidiscip Optim* 2024;67:1–24. <https://doi.org/10.1007/S00158-024-03811-0>.
- [9] Xu P, Guo W, Yang L, Yang C, Zhou S. Crashworthiness analysis and multi-objective optimization of a novel metal/CFRP hybrid friction structures. *Struct Multidiscip Optim* 2024;67:1–24. <https://doi.org/10.1007/S00158-024-03795-X/TABLES/12>.
- [10] Zhu G, Zhao Z, Wang Z, Wei L, Zhao X. Crashworthiness analysis and design optimization of square aluminum/CFRP hybrid structures under quasi-static axial loading. *Struct Multidiscip Optim* 2024;67:1–24. <https://doi.org/10.1007/S00158-024-03743-9/METRICS>.
- [11] Maria M. Advanced composite materials of the future in aerospace industry. *INCAS Bull* 2013;5:139–50. <https://doi.org/10.13111/2066-8201.2013.5.3.14>.
- [12] Ergene B, Bolat C, Karakilinc U, Irez AB. A comprehensive investigation of drilling performance of anisotropic stacked glass-carbon fiber reinforced hybrid laminate composites. *Polym Compos* 2023;44:2656–70. <https://doi.org/10.1002/PC.27268>.
- [13] Jayakrishna K, Kar VR, Sultan MTH, Rajesh M. Materials selection for aerospace components. *Sustain Compos Aerosp Appl* 2018;1–18. <https://doi.org/10.1016/B978-0-08-102131-6.00001-3>.
- [14] Kesarwani Shivi. Polymer composites in aviation sector. *Int J Eng Res* 2017;V6. <https://doi.org/10.17577/IJERTV6IS060291>.
- [15] Belov DA, Makarenko IV, Dunaev AV, Babkin AV, Solopchenko AV, Yablokova MY, et al. Curing processes simulation of complex shape carbon fiber reinforced composite components produced by vacuum infusion. *Polym Compos* 2016;37:2252–9. <https://doi.org/10.1002/PC.23403>.
- [16] Chen J, Yao W, Lin H. Compressive fatigue response and reliability analysis of thermoplastic composite with low-velocity impact damage. *Polym Compos* 2021;42:5678–90. <https://doi.org/10.1002/PC.26279>.
- [17] Vargas-Rojas E. Review of the design process of a representative test specimen of a structural joint of a carbon-epoxy aircraft. *Polym Compos* 2019;40: E898–911. <https://doi.org/10.1002/PC.25079>.
- [18] Rehman MM, Shaker K, Nawab Y. Effect of poly ether ether ketone particles on v-notched shear and drop weight impact behavior of carbon/epoxy composite. *Polym Compos* 2022;43:3219–27. <https://doi.org/10.1002/PC.26612>.
- [19] Shabeer K. Optimization of aircraft wing with composite material. *International J Innov Res Sci Eng* and 2013.
- [20] Mary N. Ply orientation of carbon fiber reinforced aircraft wing-A parametric study. *J Eng Res Appl WwwwljeraCom* 2014;4:53–5.
- [21] Zhou M, Fleury R, Kemp M. Optimization of composite - recent advances and application. 13th AIAA/ISSMO Multidiscip Anal Optim Conf 2010:2010. <https://doi.org/10.2514/6.2010-9272>.
- [22] Aung PW, Tatarnikov O, Aung NL. Structural optimization of a light aircraft composite wing. *IOP Conf Ser Mater Sci Eng* 2020;709:044094. <https://doi.org/10.1088/1757-899X/709/4/044094>.
- [23] Celik E, Sacmaozu G, Irez AB. Development of carbon-glass fiber reinforced hybrid composites: applications in offshore wind turbine blades. *Conf Proc Soc Exp Mech Ser* 2022;17–22. https://doi.org/10.1007/978-3-030-86741-6_4/FIGURES/3.
- [24] Yang W, Yue Z, Li L, Wang P. Aircraft wing structural design optimization based on automated finite element modelling and ground structure approach. *Eng Optim* 2016;48:94–114. <https://doi.org/10.1080/0305215X.2014.995175>.
- [25] Zhao W, Kapania RK. Bilevel programming weight minimization of composite flying-wing aircraft with curvilinear spars and ribs, vol. 57; 2019. p. 2594–608. <https://doi.org/10.2514/1.J057892>.
- [26] Fu Q. Optimization of a composite wing subject to multi constraints. Cranfield University; 2009.
- [27] Castro SGP, Hernandez JA. Integrating automatic zone modeling with GA in a two-step approach for structural optimization of a composite wing. 2012.
- [28] Wang Z, Luo Q, Li Q, Sun G. Design optimization of bioinspired helicoidal CFRPP/GFRPP hybrid composites for multiple low-velocity impact loads. *Int J Mech Sci* 2022;219:107064. <https://doi.org/10.1016/J.IJMECS.2022.107064>.
- [29] Shrivastava S, Mohite PM, Yadav T, Malagaudanavar A. Multi-objective multi-laminate design and optimization of a Carbon Fibre Composite wing torsion box using evolutionary algorithm. *Compos Struct* 2018;185:132–47. <https://doi.org/10.1016/J.COMPSTRUCT.2017.10.041>.
- [30] DowAksa. DowAksa carbon wrap UDS300. <https://www.s3.com.tr/uploads/GenelDosya/dowaksa-uds300-tds-823.pdf>. [Accessed 20 January 2025].
- [31] Hexion. MGS LR285 safety data sheet. <https://swiss-composite.ch/pdf/s-EPiKOTE-Resin-MGS-LR-285-e.pdf>. [Accessed 20 January 2025].
- [32] Åström BT. Manufacturing of polymer composites. Chapman & Hall; 1997.
- [33] Sidwell DR. Hand lay-up and bag molding. *Handb Compos* 1998:352–77. https://doi.org/10.1007/978-1-4615-6389-1_17.
- [34] ASTM International. D3039/D3039M standard test method for tensile properties of polymer matrix composite materials. ASTM International; 2017. https://doi.org/10.1520/D3039_D3039M-17.
- [35] ASTM International. In: D6641/D6641M standard test method for compressive properties of polymer matrix composite materials using a combined loading compression (CLC) test fixture. ASTM; 2023. https://doi.org/10.1520/D6641_D6641M-16E02.
- [36] Munk Max M, Miller EW. The aerodynamic characteristics of seven frequently used wing sections at full Reynolds number. 1927.
- [37] National Advisory Committee for Aeronautics. Annual report of the. National Advisory Committee for Aeronautics; 1915.
- [38] Tusas. Anka - TUSAS. <https://www.tusas.com/en/products/uav/operative-strategic-uav-systems/anka>. [Accessed 20 January 2025].
- [39] Air Force. MQ-1B predator > air force > fact sheet display. <https://www.af.mil/About-Us/Fact-Sheets/Display/Article/104469/mq-1b-predator/>. [Accessed 20 January 2025].
- [40] Kermode AC, Barnard RH, Philpott DR. Mechanics of flight. eleventh ed. Pearson Education.
- [41] Higgins GJ. The comparison of well-known and new wing sections tested in the variable density wind tunnel. 1925.
- [42] Ha G, Park HC. Aerodynamic and dynamic characteristics of flapping wings under low air density. *Acta Astronaut* 2024;222:692–704. <https://doi.org/10.1016/J.ACTASTRO.2024.06.048>.
- [43] Matsushima K, Shimizu R, Goshima K. Investigation into far-field drag calculation methods and pseudo total enthalpy generation for airplane CFD. *J Fluid Sci Technol* 2019;14:JFST0023. <https://doi.org/10.1299/JFST.2019JFST0023>.
- [44] Fan Y, Li W. Review of far-field drag decomposition methods for aircraft design. *J Aircraft* 2019;56:11–21. <https://doi.org/10.2514/1.C034781>.
- [45] Vyas S. ANSYS meshing user guide, vol. 12. ANSYS; 2010. https://www.academia.edu/27974461/ANSYS_Meshing_Users_Guide. [Accessed 14 April 2024].
- [46] Menter FR. Improved two-equation k-turbulence models for aerodynamic flows. 1992.
- [47] Athadkar M, Desai SS. Importance of the extent of far-field boundaries and of the grid topology in the CFD simulation of external flows. *Int J Mech Prod Eng* 2014. 2320–2092.
- [48] Chinvorarat S. Composite wing structure of light amphibious airplane design, optimization, and experimental testing. *Heliyon* 2021;7:e08410. <https://doi.org/10.1016/J.HELIYON.2021.E08410>.
- [49] Jaskowicz J, Stankiewicz A, Pluciński P. Three-dimensional numerical modelling of multi-layered shell structures using two-dimensional plane mesh. *Adv Eng Software* 2020;149:102840. <https://doi.org/10.1016/J.ADVENGSOFT.2020.102840>.
- [50] Li DH, Zhang F. Full extended layerwise method for the simulation of laminated composite plates and shells. *Comput Struct* 2017;187:101–13. <https://doi.org/10.1016/J.COMPSTRUC.2016.10.023>.
- [51] Neubauer M, Günther G, Füllhas K. Structural design aspects and criteria for military UAV. 2007. Munich.
- [52] Modlin CT, Zipay JJ. The 1.5 & 1.4 ultimate factors of safety for aircraft & spacecraft-history, definition and applications. 2014.
- [53] Hashin Z. Fatigue failure criteria for unidirectional fiber composites. *J Appl Mech* 1981;48:846–52. <https://doi.org/10.1115/1.3157744>.
- [54] Gündüz B. Structural optimization of the UAV composite wing-box by adaptive genetic algorithm methods. Istanbul Technical University; 2024.
- [55] Mahammad S, Gopi ES, Yogesh V. Roulette wheel selection-based computational intelligence technique to design an efficient transmission policy for

- energy harvesting sensors. 2020. p. 177–95. https://doi.org/10.1007/978-981-15-0994-0_12.
- [56] Michell M. *An introduction to genetic algorithms*. Mit Press; 1998.
- [57] Poli R, Langdon WB. Genetic programming with one-point crossover. *Soft Comput Eng Des Manuf* 1998;180–9. https://doi.org/10.1007/978-1-4471-0427-8_20.
- [58] Marsili-Libelli S, Alba PS. Adaptive mutation in genetic algorithms. *Soft Comput - A Fusion Found Methodol Appl* 2000;4:76–80. <https://doi.org/10.1007/S005000000042>.
- [59] Lahuerta F. Thickness effect in composite laminates in static and fatigue loading. Citation 2017. <https://doi.org/10.4233/uuid:7a2d4e15-e28b-4645-94d0-64b972064c87>.
- [60] Du B, Ji C, Chen P, Guo J, Sun A, Wei X. Study of tensile strength distribution based on composite materials for aeronautical engineering. *Mod Appl Sci* 2012;6:p21. <https://doi.org/10.5539/MAS.V6N5P21>.
- [61] Ploeckl M, Kuhn P, Grosser J, Wolfahrt M, Koerber H. A dynamic test methodology for analyzing the strain-rate effect on the longitudinal compressive behavior of fiber-reinforced composites. *Compos Struct* 2017;180:429–38. <https://doi.org/10.1016/j.COMPSTRUCT.2017.08.048>.